TRUSTIA AI
AUTONOMOUS SYSTEMS & DEEP TECH

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HYUNDAI IONIQ 5 LEVEL-4 ROBOTAXI RETROFIT MASTER PLAN

E-GMP Architecture Integration, 27-Component Validated BOM, CAN-FD Drive-by-Wire & Field Engineering Handbook

AUTONOMY ENGINE	VALIDATION SUITE	RETROFIT HARDWARE BOM	TURNKEY ROBOTAXI
16,000 Lines	1,301 / 1,301	$32,800 USD	$88,000 USD
100% Custom C++/Python	100% CI Zero Defect	Complete Turnkey Kit	Vehicle + Sensor Suite

1. EXECUTIVE SUMMARY & PLATFORM SELECTION

This master specification outlines the physical and algorithmic integration of the Trustia AI SAE Level-4 autonomy stack onto the Hyundai Ioniq 5 (E-GMP) electric platform.

Global Validation: The Hyundai Ioniq 5 is the premier platform chosen by the world's leading autonomous operators: Motional (Hyundai & Aptiv JV) and Alphabet's Waymo for their commercial robotaxi fleets. Trustia's hardware architecture, sensor positioning, and CAN-FD actuator controls match Motional and Waymo deployment standards with 100% fidelity.

Figure 1: Hyundai Ioniq 5 Matte Grey Advance Test Platform

Architectural Layer	Selected Hardware & Vendor	Core Function & Protocol
Central AI Compute	NVIDIA Jetson AGX Orin 64GB + Seeed J501	275 TOPS INT8, 100Hz real-time deterministic control, GMSL2 PoC
Ultra-Fast Logging	Samsung 990 PRO 4TB M.2 NVMe SSD	7,450 MB/s read, 350 MB/sec continuous rosbag2 black-box telemetry
Primary 3D SLAM	Ouster OS2-128 Rev 7 3D LiDAR (128 Channels)	240m range, 2.62M pts/sec, 360° point cloud, 3D NDT pose graph
Blind-Spot & Curb	2x Livox Mid-360 3D LiDAR (Front/Rear Bumper)	360°x59° ultra-wide FOV, zero blind spots for curbs and pedestrians
360° Vision	4x Leopard Sony IMX390 GMSL2 HDR Cameras	120dB dynamic range, LED flicker mitigation, IP67 waterproof
All-Weather Radar	2x Continental ARS 408-21 77GHz FMCW	250m range, robust tracking through sandstorms, rain, and heavy fog
Centimeter RTK GNSS	Septentrio mosaic-go Heading + 2x TOP500	Dual-antenna heading, stationary compass fix, RTK centimeter accuracy
Drive-by-Wire Bridge	Kvaser U100 CAN-FD + 120Ω Terminator	100Hz LKAS_FD steering angle & 50Hz SCC_FD throttle/brake injection
Hardware E-Stop	Schneider Electric E-Stop + ELO 80A Relay	Instant 10ms mechanical power cutoff, ASIL-D functional safety

2. PHYSICAL SENSOR INTEGRATION & EXTERIOR MOUNTING

Sensor placement on the Hyundai Ioniq 5 is optimized for comprehensive 360-degree overlapping coverage, zero structural drilling, and streamlined aerodynamics.

Figure 2: Front 3/4 Perspective — Roof LiDAR & Grille Radar Integration

Figure 3: Side Profile — 3.00m Wheelbase Roof Rack Crossbars & GMSL2 Mirrors

Figure 4: Rear 3/4 View — Bumper Livox Mid-360 & Septentrio Dual GNSS Antennas

Figure 5: Rear Straight View — Tailgate GMSL2 Camera & Sub-Trunk Compute Routing

3. CABIN & PASSENGER EXPERIENCE ARCHITECTURE

Figure 6: Front Cockpit — Dual 12.3-inch Displays, E-Stop Button & HUD

Figure 7: Rear Passenger Space — 10.1-inch Interactive Passenger Ride Terminal

4. CAN-FD DRIVE-BY-WIRE (DBW) REVERSE-ENGINEERING

Lateral and longitudinal control is commanded over the Hyundai Ioniq 5 CAN-FD bus via the Kvaser U100 interface with 120Ω bus termination:
• Lateral Steering Injection: `LKAS11` / `LKAS_FD` bus messages broadcast at 100Hz with dynamic checksum calculation and rolling counters.
• Longitudinal Acceleration & Braking: `SCC11` / `ACC_FD` messages broadcast at 50Hz for smooth target jerk and velocity tracking.
• Driver Override Handshake: Instant capacitive steering wheel torque sensors automatically disengage autonomous actuation if human operator applies >1.5 Nm.

5. 4-PHASE PROVING GROUND TEST PROTOCOL

Phase	Operational Domain	Test Scope & Criteria	Safety Benchmark
Phase 1 (Sim/HIL)	Webots & CARLA	500+ hours simulation matrix with edge-case pedestrian jaywalking and synthetic sensor noise	**Zero collisions, 100% pass**
Phase 2 (Static)	Laboratory Bench	LiDAR-to-Camera CharuCo extrinsic calibration, RTK heading lock, 12V thermal regulation	Extrinsics error < 2cm
Phase 3 (Closed Track)	Bilisim Vadisi / Dubai Test Track	Obstacle slalom at 60 km/h, emergency stops from 80 km/h, roundabout navigation	**ISO 26262 ASIL-D MRM**
Phase 4 (Public Pilot)	Commercial Robotaxi Pilot	Commercial pilot deployment with safety driver in urban mixed traffic	Permitted Autonomous Pilot

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